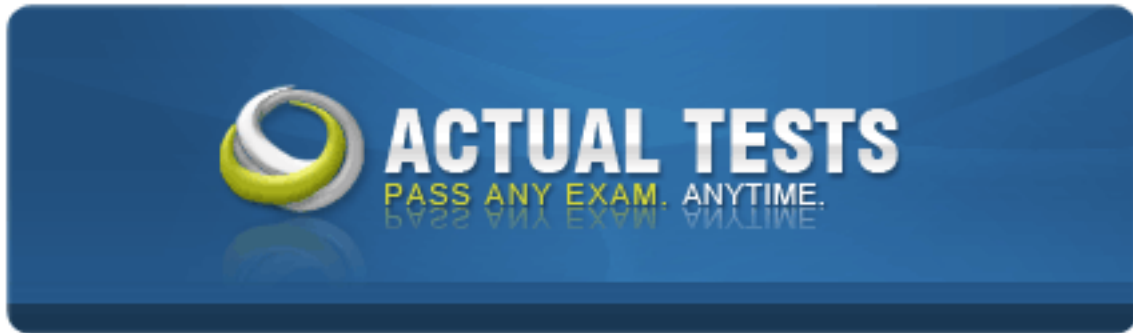


Cisco 642-887



Implementing Cisco Service Provider Next-Generation Core Network Services

Version: 4.1

QUESTION NO: 1

Which three conditions can occur when metering traffic using a dual token bucket traffic policing QoS mechanism on Cisco routers? (Choose three.)

- A. conform
- B. pass
- C. violate
- D. exceed
- E. burst
- F. matched

Answer: A,C,D

Explanation:

QUESTION NO: 2

What is the correct formula for determining the CIR?

- A. $CIR = Bc/Tc$
- B. $CIR = Bc \times Tc$
- C. $CIR = Tc/Bc$
- D. $CIR = Bc + Be$
- E. $CIR = Tc/(Bc+Be)$
- F. $CIR = (Bc+Be)/Tc$

Answer: A

Explanation:

QUESTION NO: 3

DS-TE implementations on Cisco routers support which bandwidth pool(s) and class type(s)? (Choose two.)

- A. global pool only
- B. subpool only
- C. global pool and subpool
- D. class-type 0 only
- E. class-type 1 only
- F. class-type 0 and class-type 1

Answer: C,F

Explanation:

QUESTION NO: 4

Which field in the MPLS shim header is used to support different QoS markings?

- A. IP precedence
- B. DSCP
- C. EXP
- D. ToS
- E. S
- F. Label

Answer: C

Explanation:

QUESTION NO: 5

On a Cisco IOS XR router, which mechanism protects the router resources by filtering and policing the packets flows that are destined to the router that is based on defined flow-type rates?

- A. LLQ
- B. LPTS
- C. Committed Access Rate
- D. Control Plane Policing
- E. Management Plane Protection
- F. NetFlow
- G. ACL

Answer: B

Explanation:

QUESTION NO: 6

When configuring LLQ (strict priority queue) on a traffic class using the Cisco IOS XR priority command on a Cisco ASR9K router, which additional QoS command is required for this traffic class?

- A. shape
- B. police
- C. random-detect
- D. bandwidth

Answer: B

Explanation:

QUESTION NO: 7

On the Cisco ASR9K router, when using the bandwidth command to specify the minimum guaranteed bandwidth to be allocated for a specific class of traffic, what will be used as the queuing algorithm?

- A. custom queuing
- B. CBWFQ
- C. WFQ
- D. FIFO
- E. priority queuing

Answer: B

Explanation:

QUESTION NO: 8

When implementing MPLS DS-TE on Cisco IOS XR routers, all aggregate Cisco MPLS TE traffic is mapped to which class type by default?

- A. class-type 0 (bandwidth global pool)
- B. class-type 1 (bandwidth subpool)
- C. class-type 2 (bandwidth priority)
- D. class type class-default (bandwidth best-effort)

Answer: A

Explanation:

QUESTION NO: 9

On the Cisco IOS XR, which MQC configuration is different than on the Cisco IOS and IOS XE?

- A. On the Cisco IOS XR, WRED can only be applied in the output direction.
- B. On the Cisco IOS XR, marking can only be applied in the input direction.
- C. On the Cisco IOS XR, LLQ can be applied in the input or output direction.
- D. On the Cisco IOS XR, LLQ can use up to four priority queues: level 1, level 2, level 3, and level 4.

Answer: C

Explanation:

QUESTION NO: 10

On Cisco routers, how is hierarchical QoS implemented?

- A. Within the parent policy, reference another child policy using the policy-map command.
- B. Within the child policy, reference another parent policy using the policy-map command.
- C. Use the policy-map command within a service-policy to implement nested policy-maps.
- D. Within the parent policy-map, reference another child policy-map using the service-policy command.

Answer: D

Explanation:

QUESTION NO: 11

Refer to the Cisco IOS XR policy-map configuration exhibit.

```
policy-map test
!
class one
priority level 1
!
class two
priority level 2
!
class three
bandwidth percent 60
!
interface GigabitEthernet0/0/0/2
service-policy output test
!
!
```

What is wrong with the policy-map configuration?

- A. missing the priority percent command under class one and class two
- B. missing the police command under class one and class two
- C. missing the police command under class three
- D. missing the priority bandwidth command under class one and class two
- E. missing the bandwidth command under class one and class two

Answer: B

Explanation:

QUESTION NO: 12

When configuring class-based WRED on Cisco routers, which WRED parameter is not user configurable on a Cisco IOS XR but is user configurable on a Cisco IOS and IOS XE?

- A. the ingress or egress direction where the class-based WRED policy will be applied
- B. the maximum threshold
- C. the minimum threshold
- D. the mark probability denominator

Answer: D

Explanation:

QUESTION NO: 13

Which of the following three statements are correct regarding IPv6 QoS? (Choose three.)

- A. The traffic class field in the IPv6 header can be used to set specific precedence or DSCP values.
- B. A 20-bit flow label field enables per-flow processing.
- C. DS-TE is not supported by IPv6.
- D. Per-hop behavior in IPv6 networks is based on EXP bits.
- E. IPv6 QoS features are configured using the modular QoS CLI on Cisco routers.

Answer: A,B,E

Explanation:

QUESTION NO: 14

With unmanaged CE routers, at which point in the service provider network is the QoS trust boundary, and what is required at the trust boundary?

- A.** between the CE and PE router and mapping of the customer traffic classes into the service provider traffic classes at the PE router ingress
- B.** between the CE and PE router and trusting the QoS markings from the CE router and applying the required QoS mechanisms based on the customer QoS markings
- C.** between the PE and the P router and mapping of the customer traffic classes into the service provider traffic classes at the P router ingress
- D.** between the PE and P router and trusting the QoS markings from the CE router and applying the required QoS mechanisms based on the customer QoS markings
- E.** between the customer network and the CE router ingress and applying the required egress QoS policy on the CE router

Answer: A

Explanation:

QUESTION NO: 15

On the Cisco IOS XR, when using the match protocol command within a class-map to classify traffic, you noticed that the match protocol option on the Cisco IOS XR shows much fewer protocol options than on the Cisco IOS or IOS XE, like there is no option such as the match protocol yahoo-messenger command on the Cisco IOS XR. Why is this?

- A.** because the Cisco IOS XR router does not have the correct software packages installed
- B.** because when defining the class-map, the class-map type should be set to type inspect: class-map type inspect class-map-name command
- C.** because NBAR is not supported on the Cisco IOS XR
- D.** because flexible packet matching has not been enabled on the Cisco IOS XR router

Answer: C

Explanation:

QUESTION NO: 16

Within the service provider core network, which two QoS mechanisms are typically deployed on the P routers? (Choose two.)

- A. LLQ
- B. traffic policing and remarking
- C. WRED
- D. traffic shaping
- E. traffic classification and markings
- F. link fragmentation and interleaving

Answer: A,C

Explanation:

QUESTION NO: 17

Which three steps are required to configure QPPB on Cisco IOS XR routers? (Choose three.)

- A. Apply a QPPB route policy to the BGP process using the table-policy command
- B. Apply a QPPB route policy to the BGP neighbor using the route-policy command
- C. Define a QPPB route policy to match the customer routes, then set the IP precedence or qos-group
- D. Define a QPPB route policy to match the customer IP precedence or qos-group markings, then set the BGP community
- E. Enable QPPB on an interface using the ipv4 bgp policy propagation input ip-precedence|qos-group destination|source command
- F. Enable QPPB on an interface using the ipv4 bgp policy propagation output ip-precedence|qos-group destination|source command

Answer: A,C,E

Explanation:

QUESTION NO: 18

The Cisco IOS and IOS XE qos pre-classify command allows which kind of packet classification on IP packets that are encapsulated with GRE and IPsec?

- A. allows for packets to be classified based on the ToS byte values before packet encryption
- B. allows for packets to be classified based on the ToS byte values after packet encryption
- C. allows for packets to be classified based on the packet payload before packet encryption
- D. allows for packets to be classified based on the packet payload after packet encryption
- E. allows for packets to be classified based on the packet header parameters other than the ToS byte values after packet encryption

Answer: E

Explanation:

QUESTION NO: 19

Which are typical class-based marking policies that are implemented on service provider IP NGN PE routers?

- A.** On the PE ingress, classify the customer traffic and then mark with qos-group. On the PE egress, classify based on the qos-group and then mark with mpls exp.
- B.** On the PE ingress, classify the customer traffic and then mark with mpls exp. On the PE egress, classify based on the mpls exp and then mark with qos-group.
- C.** On the PE ingress, trust the customer QoS markings. On the PE egress, classify based on the customer QoS markings and then mark with qos-group.
- D.** On the PE ingress, trust the customer QoS markings. On the PE egress, classify based on the customer QoS markings and then mark with mpls exp.

Answer: A

Explanation:

QUESTION NO: 20

Which item is not available to be used for QoS classification in Cisco IOS XR?

- A.** MAC SA
- B.** protocol
- C.** inner EXP
- D.** discard-class
- E.** QoS-group
- F.** VLAN

Answer: C

Explanation:

QUESTION NO: 21

Which statement is correct regarding the default MPLS TTL behavior?

- A.** When an ingress edge LSR receives an IP packet, it will decrement the IP TTL field by 1; then it

will set the MPLS Label TTL field to 255.

B. When an ingress edge LSR receives an IP packet, it will decrement the IP TTL field by 1; then it will copy the decremented IP TTL field into the MPLS Label TTL field.

C. When an ingress edge LSR receives an IP packet, it will just copy the IP TTL field into the MPLS Label TTL field.

D. When an ingress edge LSR receives an IP packet, it will copy the IP TTL field into the MPLS Label TTL field first; then it will only decrement the MPLS Label TTL field by 1.

E. When an ingress edge LSR receives an IP packet, it will copy the IP TTL field into the MPLS Label TTL field first; then it will only decrement the IP TTL field by 1.

Answer: B

Explanation:

QUESTION NO: 22

Which three statements are correct regarding ping mpls command operations? (Choose three.)

A. MPLS OAM has to be enabled on the router using the mpls oam command.

B. They use a 127/8 address as the destination address in the MPLS echo request packet.

C. They use ICMP echo request and ICMP echo reply packets.

D. They are used to test for a broken LSP.

E. If there is a broken LSP, instead of using label switching, the packet can still be forwarded based on the destination IP address in the mpls ping echo request packet.

Answer: A,B,D

Explanation:

QUESTION NO: 23

What are the four fields inside the MPLS shim header? (Choose four.)

A. EXP

B. TTL

C. Version

D. S

E. Length

F. Label

G. Type

H. FCS

Answer: A,B,D,F

Explanation:

QUESTION NO: 24

Which two of the following statements are correct regarding LSPs? (Choose two.)

- A. An LSP is created for every routing protocol entry.
- B. Each LSP is bidirectional, that is, packets traveling in the opposite direction use the same LSP.
- C. An IGP is used to populate routing tables in all routers in an MPLS domain.
- D. LDP is used to propagate labels and build LSPs.
- E. The FIB is used to forward MPLS-labeled packets down an LSP.

Answer: C,D

Explanation:

QUESTION NO: 25

Which four pieces of information are stored for each prefix in the LFIB? (Choose four.)

- A. local label
- B. outgoing label
- C. next-hop IP address
- D. outgoing interface
- E. incoming interface
- F. Layer 2 header rewrite information

Answer: A,B,C,D

Explanation:

QUESTION NO: 26

Which three network services can be implemented using MPLS within the service provider IP NGN core? (Choose three.)

- A. Layer 2 VPNs
- B. Layer 3 VPNs
- C. traffic engineering
- D. IntServ traffic engineering tunnels

E. encrypted LSPs

Answer: A,B,C

Explanation:

QUESTION NO: 27

LDP session protection uses which one to maintain the LDP session between LDP neighbors?

- A. LDP NSF
- B. LDP NSR
- C. backup-targeted LDP hellos
- D. BFD
- E. LDP-IGP synchronization

Answer: C

Explanation:

QUESTION NO: 28

You are tasked to enable LDP on many of the interfaces on the Cisco CRS-3 router, and because there are many interfaces that need to have LDP enabled, you mistakenly did not enable LDP on all the required interfaces. To prevent this issue from happening again in the future, what could you do the next time you need to enable LDP on many interfaces?

- A. use the mpls ldp auto-config command under the IGP routing process
- B. use the mpls ldp sync command under the IGP routing process
- C. use the interface all command under the MPLS LDP process
- D. use the discovery command under the MPLS LDP process

Answer: A

Explanation:

QUESTION NO: 29

Which two statements are correct in describing ISP environments that are running IP/MPLS in the core network? (Choose two.)

- A. On the PE routers, each BGP route must use a unique label to reach the BGP next hop.
- B. The BGP next hops point to the PE routers, and only the PE routers are required to run BGP.
- C. A full mesh of IBGP sessions are required between all of the PE and P routers to ensure proper packets forwarding.
- D. The PE and P routers run LDP to learn the labels for reaching the BGP next-hop addresses.

Answer: B,D

Explanation:

QUESTION NO: 30

When troubleshooting LDP operations on the Cisco IOS and IOS XE routers, what is one of the first things that should be verified?

- A. if running OSPF as the IGP, ensure that OSPFv3 has been enabled
- B. check if the ip cef command has been enabled
- C. verify in the running configurations that all of the required LDP interfaces are defined under the mpls ldp command configuration mode
- D. verify if there are any access lists that are denying TCP and UDP port 464 packets

Answer: B

Explanation:

QUESTION NO: 31

Referring to the Cisco IOS XR configuration exhibit,

```
mpls ldp
label
advertise
disable
for test1 to test2
!
ipv4 access-list test2
10 permit ipv4 any any
ipv4 access-list test1
10 permit ipv4 host 10.1.1.1 any
```

which labels will be advertised by the router with this configuration?

- A. Only the label for 10.1.1.1/32 will be advertised to all the LDP peers.
- B. Labels for all prefixes will be advertised to the 10.1.1.1 LDP peer.
- C. Labels for all prefixes will be advertised to all the LDP peers.
- D. No labels will be advertised to any LDP peers.

Answer: A

Explanation:

QUESTION NO: 32

What is the term that is used for the label that an LSR assigns and distributes to other LSRs in MPLS?

- A. Local
- B. Remote
- C. Explicit
- D. Explicit Null
- E. Aggregate

Answer: A

Explanation:

QUESTION NO: 33

Which Cisco IOS XR high-availability feature is used to prevent routes from being used before LDP converges?

- A. LDP session protection
- B. LDP-IGP synchronization
- C. BFD
- D. IGP session protection

Answer: B

Explanation:

QUESTION NO: 34

Which Cisco IOS XR command should be used in order to enable LDP-IGP synchronization for

the ISIS IGP protocol?

- A. RP/0/RSP0/CPU0:R1(config-isis-if-af)#mpls ldp sync
- B. RP/0/RSP0/CPU0:R1(config-isis)#mpls ldp sync
- C. RP/0/RSP0/CPU0:R1(config-ldp)#isis ldp sync
- D. LDP-IGP synchronization is not supported for the ISIS IGP protocol on Cisco IOS XR platforms

Answer: A

Explanation:

QUESTION NO: 35

Referring to the Cisco IOS XR show command output exhibit,

```
RP/0/RSP1/CPU0:ASR9006#sh mpls ldp igp sync
Bundle-Ether9000:
  Sync status: Ready
  Peers:
    192.168.1.25:0 (GR)
GigabitEthernet0/1/0/10:
  Sync status: Not ready
```

what are three possible reasons that the GigabitEthernet0/1/0/10 LDP IGP sync status is not ready? (Choose three.)

- A. GigabitEthernet0/1/0/10 is not configured to run LDP.
- B. Graceful restart is not configured on the peer.
- C. The LDP neighbor on GigabitEthernet0/1/0/10 is not up.
- D. The OSPF neighbor on GigabitEthernet0/1/0/10 is not up.
- E. LDP is up on GigabitEthernet0/1/0/10, but no label bindings have been received from the peer.
- F. GigabitEthernet0/1/0/10 is a member link of Bundle-Ether9000.

Answer: C,D,E

Explanation:

QUESTION NO: 36

In which two Cisco IOS XR configuration modes can mpls ldp igp sync be configured? (Choose

two.)

- A. config-ldp
- B. config-if
- C. config-ospf-ar
- D. config-ospf
- E. config-isis

Answer: C,D

Explanation:

QUESTION NO: 37

On Cisco routers, which three methods can be used to map traffic into the MPLS traffic engineering tunnel? (Choose three.)

- A. on-demand routing
- B. static routing
- C. optimized edge routing
- D. policy-based routing
- E. autoroute

Answer: B,D,E

Explanation:

QUESTION NO: 38

Which three statements are correct regarding a Cisco MPLS TE? (Choose three.)

- A. A Cisco MPLS TE tunnel maps onto an LSP path.
- B. Tunnels are bidirectional by default.
- C. Packets that are mapped into a Cisco MPLS TE tunnel will have two labels, with the top label indicating what the tail-end router should do with the packet.
- D. A tunnel that is created with a priority of 0 can pre-empt an existing tunnel with a priority of 7.
- E. CBR takes into account link resource and traffic tunnel attributes.
- F. RSVP is used between customer routers.

Answer: A,D,E

Explanation:

QUESTION NO: 39

On a Cisco router, when will the router actually reserve the bandwidth for the MPLS traffic engineering tunnel?

- A. during the autoroute process
- B. during constraint-based routing calculations
- C. on the receipt of the RSVP Path message
- D. on the receipt of the RSVP Resv message

Answer: D

Explanation:

QUESTION NO: 40

Cisco MPLS TE path setup can be affected by which three tunnel attributes? (Choose three.)

- A. bandwidth
- B. delay
- C. MTU
- D. priority
- E. affinity

Answer: A,D,E

Explanation:

QUESTION NO: 41

When using the tunnel mpls traffic-eng path-option 1 explicit name test command in Cisco MPLS TE tunnel configurations, the test explicit-path configuration will consist of a list of which values?

- A. tunnel endpoint
- B. resource class affinity
- C. MPLS label
- D. IP address

Answer: D

Explanation:

QUESTION NO: 42

Cisco MPLS TE resource attributes that are configured locally for each link are distributed to the headend router of the traffic engineering tunnel using which protocol?

- A. BGP
- B. MP-BGP
- C. LDP
- D. RSVP
- E. OSPF or IS-IS with TE extension

Answer: E

Explanation:

QUESTION NO: 43

When implementing Cisco MPLS TE, the constrained-based path calculations will use which value as the TE cost of each link within the MPLS domain?

- A. By default, it will use the IGP metric, or each link can be assigned a specific value using the admin-weight command.
- B. It can only use the IGP metric as the TE cost.
- C. It will use the interface bandwidth as the TE cost.
- D. Each link must be assigned a TE cost using the metric command.

Answer: A

Explanation:

QUESTION NO: 44

Which affinity and mask value will match the link affinity that has 0x0F in the first 8 bits and 0x01 in the last 8 bits, and the middle 16 bits can be any value?

- A. affinity 0x0F000001 mask 0x0F000001
- B. affinity 0x0F000001 mask 0x00000000
- C. affinity 0x0F000001 mask 0x11111111

- D. affinity 0x0F000001 mask 0x11000011
- E. affinity 0x0F000001 mask 0xFF0000FF
- F. affinity 0x0F000001 mask 0x00FFFF00

Answer: E

Explanation:

QUESTION NO: 45

Which two features are used to provide Cisco MPLS TE node and link protection? (Choose two.)

- A. autoroute
- B. fast reroute
- C. backup tunnels
- D. BFD

Answer: B,C

Explanation:

QUESTION NO: 46

In Cisco MPLS TE implementations, what can cause the tunnel bandwidth to adjust automatically based on the traffic load in the tunnel?

- A. fast reroute
- B. admin weight
- C. autobandwidth
- D. bandwidth subpool

Answer: C

Explanation:

QUESTION NO: 47

Which affinity value will be matched by the affinity bit mask of the affinity 0xFF00000A mask 0xFFFF000F command?

- A. 0xFFFF000A

- B. 0xFF00EEEEA
- C. 0xFF000000
- D. 0x00000000
- E. 0x11110001

Answer: B

Explanation:

QUESTION NO: 48

Refer to the Cisco IOS XR configuration exhibit.

```
interface Tunnel-te 10
ipv4 unnumbered Loopback0
destination 10.5.5.5
signalled-bandwidth 1000
priority 7 7
path-option 1 explicit name testpath
!
explicit-path name testpath
index 1 next-address ipv4 unicast 10.3.3.3
index 2 next-address ipv4 unicast 10.4.4.4
!
mpls traffic-eng
interface GigabitEthernet 0/0/0/10
backup-path tunnel-te 10
```

Which statement is correct?

- A. The backup tunnel-te 10 tunnel is using the highest setup and hold priority settings of 7.
- B. The backup tunnel path is learned dynamically.
- C. The fast-reroute command is missing under the (config-mpls-te-if)# configuration mode.
- D. Interface gi0/0/0/10 is the protected link.

Answer: D

Explanation:

QUESTION NO: 49

When defining an explicit MPLS TE tunnel path, which two command options are available under

the explicit-path configuration mode? (Choose two.)

- A. exclude-address
- B. include-address
- C. next-address
- D. dynamic-address

Answer: A,C

Explanation:

QUESTION NO: 50

Which three mechanisms are used to implement MPLS TE? (Choose three.)

- A. tunnel interface
- B. CSPF
- C. RSVP
- D. LDP
- E. MP-BGP

Answer: A,B,C

Explanation:

QUESTION NO: 51

Implementing IPoDWDM interfaces on Cisco CRS routers eliminates the need for which network component?

- A. ROADM
- B. external transponders
- C. electrical-optical-electrical converters
- D. electrical cross-connect

Answer: B

Explanation:

QUESTION NO: 52

Layer 2 VPN services that are offered by traditional service providers using a SONET/SDH backbone can be implemented by service providers using an IP/MPLS backbone with which MPLS feature?

- A. LSP stitching
- B. AToM
- C. virtual private WAN services
- D. cell-mode MPLS

Answer: B

Explanation:

QUESTION NO: 53

On Cisco routers, which QoS marker is only locally significant?

- A. DSCP
- B. MPLS EXP
- C. IP precedence
- D. QoS group
- E. discard eligible (DE)

Answer: D

Explanation:

QUESTION NO: 54

Which QoS mechanism is used for congestion avoidance?

- A. LLQ
- B. CBWFQ
- C. WRED
- D. LFI
- E. traffic policing

Answer: C

Explanation:

QUESTION NO: 55

Only based on the Cisco IOS XR policy-map configuration exhibit,

```
policy-map policy_A
class test
bandwidth 1000000
random-detect dscp AF11 10000 20000
random-detect dscp AF41 12000 24000
```

which statement is correct?

- A.** All DSCP AF41 marked packets will be dropped when the average queue length reaches 12,000 packets.
- B.** DSCP AF11 marked packets will be randomly dropped when the average queue length reaches 10,000 packets.
- C.** DSCP AF11 and AF41 marked packets are guaranteed a minimum bandwidth of 1 Mb/s.
- D.** DSCP AF11 and AF41 marked packets are guaranteed a maximum bandwidth of 1 Mb/s.

Answer: B

Explanation:

QUESTION NO: 56

Referring to the show output exhibit,

```
RP/0/RSP0/CPU0:P1# show mpls ldp bindings
```

```
10.7.1.1/32, rev 61
```

```
Local binding: label: 16008
```

```
Remote bindings: (2 peers)
```

```
Peer          Label
```

```
-----
```

```
10.0.2.1:0    16013
```

```
10.7.1.1:0    IMP-NULL
```

```
10.7.10.1/32, rev 85
```

```
Local binding: label: 16009
```

```
Remote bindings: (1 peer)
```

```
Peer          Label
```

```
-----
```

```
10.0.2.1:0    16022
```

which statement is correct?

- A. The P1 router is receiving label information for the 10.0.2.1/32 prefix from two LDP peers.
- B. The P1 router is receiving label information for the 10.7.1.1/32 prefix from two LDP peers.
- C. The P1 router is using the IMP-NULL local label for the 10.7.1.1/32 prefix because 10.7.1.1/32 is the loopback interface IP address on the P1 router.
- D. The P1 router will perform PHP for the 10.7.1.1/32 and 10.7.10.1/32 prefixes.

Answer: B

Explanation:

QUESTION NO: 57

Referring to the traceroute output exhibit that is shown,

```
pe1#traceroute 14.14.14.14
```

```
Type escape sequence to abort.
```

```
Tracing the route to 14.14.14.14
```

```
1 37.37.37.1 [MPLS: Label 66 Exp 0] 40 msec 24 msec 28 msec
2 78.78.78.2 [MPLS: Label 99 Exp 0] 28 msec 32 msec 28 msec
3 181.181.181.1 [MPLS: Label 99 Exp 0] 36 msec 24 msec 24 msec
4 110.110.110.1 28 msec 28 msec 28 msec
5 103.103.103.2 [MPLS: Label 66 Exp 0] 28 msec 28 msec 24 msec
6 135.135.135.2 28 msec 28 msec *
```


which statement is correct?

- A. There is no problem with the end-to-end LSP as indicated by the successful trace.
- B. Normal PHP operation is performed by the hop 4 router.
- C. The end-to-end LSP is broken at hop 4.
- D. At each hop, each LSR is able to perform label swapping.

Answer: C

Explanation:

QUESTION NO: 58

What is an important requirement with MPLS applications like Layer 3 MPLS VPNs?

- A. All the PE routers loopback addresses should be summarized to reduce the number of routing table entries in the core routers.
- B. Targeted hellos are required between all the PE routers.
- C. An end-to-end LSP is required between the PE routers.
- D. The LSPs that are built between the PE routers must be symmetrical (bidirectional).

Answer: C

Explanation:

QUESTION NO: 59

When implementing MPLS TE tunnels on Cisco IOS XR routers, what is the tunnel setup and hold priority value range, and which value has the highest priority?

- A. 0-63, where 0 is the highest priority
- B. 0-63, where 63 is the highest priority
- C. 0-7, where 0 is the highest priority
- D. 0-7, where 7 is the highest priority

Answer: C

Explanation:

QUESTION NO: 60

Which Cisco IOS XR command should be used in order to enable LDP on all interfaces for which the IGP protocol is enabled?

- A. RP/0/0/CPU0:R1(config-ospf)#mpls ldp auto-config
- B. RP/0/0/CPU0:R1(config-ospf)#mpls ldp interface all enable
- C. RP/0/0/CPU0:R1(config-ospf)#enable all
- D. RP/0/0/CPU0:R1(config-ldp)#enable all

Answer: A

Explanation:

QUESTION NO: 61

Refer to the partial Cisco IOS XR configurations exhibit for Router 1 and Router 2.

```
RP/0/RP0/CPU0:Router1#show running-config mpls ldp
```

```
mpls ldp
router-id 33.33.33.33
log
adjacency
!
interface GigabitEthernet0/2/0/6
!
```

```
RP/0/RP1/CPU0:Router2#show running-config mpls ldp
```

```
mpls ldp
router-id 10.12.0.3
log
neighbor
!
interface GigabitEthernet0/0/2/2
!
```

```
RP/0/RP0/CPU0:Router1#show running-config router ospf
```

```
router ospf test
area 0
interface Loopback7
passive enable
!
```

```
RP/0/RP1/CPU0:Router2#show running-config router ospf
```

```
router ospf test
area 0
interface Loopback0
passive enable
!
interface GigabitEthernet0/0/2/2
!
```

There are two routers that are connected back to back over the Gigabit Ethernet link. If the "show mpls ldp neighbor" command output on Router 1 does not show LDP peering with Router 2, what could be the possible root cause of the LDP peering problem?

A. missing interface under OSPF IGP configuration

- B. hello timers mismatch on Router 1 and Router 2
- C. password for LDP session mismatch on Router 1 and Router 2
- D. MPLS LDP session protection is not configured

Answer: A

Explanation:

QUESTION NO: 62

Which Cisco IOS XR command should be used to identify if MPLS TE FRR is enabled?

- A. show mpls traffic-eng tunnel <tunnel#>
- B. show mpls frr
- C. show mpls traffic-eng protection
- D. show mpls protection
- E. show mpls fast-reroute

Answer: A

Explanation:

QUESTION NO: 63

When a link flaps, an mpls ldp session will also flap. Which feature minimizes traffic loss and provides faster convergence after the link is re-established?

- A. BFD
- B. MPLS LDP IGP SYNC
- C. graceful restart
- D. LDP session protection
- E. LDP nonstop routing

Answer: D

Explanation:

QUESTION NO: 64

A DSCP value of 41 in decimal corresponds to which IP precedence value?

- A. 3 – Flash
- B. 4 – Flash Override
- C. 5 – Critical
- D. 6 – Internet Control
- E. 7 – Network Control

Answer: C

Explanation:

QUESTION NO: 65 DRAG DROP

Referring to Cisco MPLS TE path setup operations using RSVP, drag the RSVP characteristic on the left to match the correct RSVP message type on the right.

sent from the tunnel headend to the tunnel tailend	RSVP PATH Message
sent from the tunnel tailend to the tunnel headend	Target
carries the MPLS label requests	Target
carries the MPLS labels	RSVP RESV Message
	Target
	Target

Answer:

Referring to Cisco MPLS TE path setup operations using RSVP, drag the RSVP characteristic on the left to match the correct RSVP message type on the right.

sent from the tunnel headend to the tunnel tailend	RSVP PATH Message
sent from the tunnel tailend to the tunnel headend	sent from the tunnel headend to the tunnel tailend
carries the MPLS label requests	carries the MPLS label requests
carries the MPLS labels	RSVP RESV Message
	sent from the tunnel tailend to the tunnel headend
	carries the MPLS labels

Explanation:

RSVP PATH Message --- sent from the tunnel headend to the tunnel tailend

---- carried the MPLS label requests

RSVP RESV Message—Sent from the tunnel tailend to the tunnel headend

--- carries the MPLS labels

QUESTION NO: 66 DRAG DROP

Drag each of the QoS mechanisms on the left to match the correct description on the right. (Not all options on the left are used.)

LLQ	Can drop excess traffic beyond the committed rate and remark nonconforming traffic before transmitting it.
LFI	Use to avoid the TCP global synchronization problems that occur when tail drop is used as the congestion avoidance mechanism.
traffic shaping	Provide a strict priority queue to allow delay-sensitive data such as voice to be dequeued and sent first.
WRED	Excess traffic beyond the committed rate will be queued and scheduled for later transmission. Only applied in the output direction.
CBWFQ	
traffic policing	
WFQ	

Answer:

Drag each of the QoS mechanisms on the left to match the correct description on the right. (Not all options on the left are used.)

LLQ	traffic policing
LFI	WRED
traffic shaping	CBWFQ
WRED	traffic shaping
CBWFQ	
traffic policing	
WFQ	

Explanation:

Can drop express traffic beyond the committed rate and remark non coming traffic before transmitting it—traffic policing

Use to avoid the TCP global synchronization problems that occur when tail drop is used as the congestion avoidance mechanism---WRED

Provide a strict priority queue to allow delay-sensitive data such as voice t be dequeued and sent first--CBWFQ

Express traffic beyond the committed rate will be queue and scheduled for later transmission .Only applied in the output directions.—traffic shaping

QUESTION NO: 67 DRAG DROP

Drag the Cisco MQC configuration task on the left to match the correct description on the right. (Not all options on the left are required.)

class-map	applies the QoS policy to an interface
tcp-map	defines the PHB QoS action(s) for each of the different traffic classes
route-map	defines the matching parameter(s) for classifying packets into service classes
policy-map	
service-policy	
route-policy	
qos-group	

Answer:

Drag the Cisco MQC configuration task on the left to match the correct description on the right. (Not all options on the left are required.)

class-map	service-policy
tcp-map	policy-map
route-map	class-map
policy-map	
service-policy	
route-policy	
qos-group	

Explanation:

Applies the QOS policy to interface—service policy

Defines the PHB qos action for each of the different traffic classes—policy map

Defines the making parameters for classifying packets into service classes—class-map

QUESTION NO: 68 DRAG DROP

Put the MPLS LDP steps on the left into the correct order from top to bottom on the right.

Build RIB	Target
Build LFIB	Target
Assign Labels	Target
Advertise Labels	Target

Answer:

Put the MPLS LDP steps on the left into the correct order from top to bottom on the right.

Build RIB	Build RIB
Build LFIB	Assign Labels
Assign Labels	Advertise Labels
Advertise Labels	Build LFIB

Explanation:

- 1) Build RIB
- 2) Assign labels
- 3) Advertise Labels
- 4) Build LFIB

QUESTION NO: 69 DRAG DROP

Drag the QoS model on the left to match its correct description on the right.

DiffServ	All network packets are treated exactly the same.
IntServ	It divides traffic into classes and applies a different level of service for each class.
Best Effort	Traffic-handling characteristics are based on signaling events from network-based applications.

Answer:

Drag the QoS model on the left to match its correct description on the right.

DiffServ	Best Effort
IntServ	DiffServ
Best Effort	IntServ

Explanation: All network packets are treated exactly the same—Best Effort
 It divides traffic classes and applies a different level of service for each class--Diffserv
 Traffic-handling characteristics are based on signaling events from network-based applications--intServ

QUESTION NO: 70 DRAG DROP

Drag the MPLS Diff-Serv tunneling mode on the left to match the correct description on the right.

Pipe Mode	If a QoS marking (MPLS EXP) is changed in the MPLS network, it is also changed in the egress IP packet.
Short-Pipe Mode	This provides QoS transparency where the customer QoS marking (for example, DSCP) in the IP packet is preserved. The egress PE uses the original customer QoS marking instead of the service provider QoS marking.
Uniform Mode	QoS is done on the output interface of the PE router that is based on the received MPLS EXP field. The customer QoS marking (for example, DSCP) is not altered when the customer packet travels from the ingress to the egress of the MPLS domain.

Answer:

Drag the MPLS Diff-Serv tunneling mode on the left to match the correct description on the right.

Pipe Mode	Short-Pipe Mode
Short-Pipe Mode	Pipe Mode
Uniform Mode	Uniform Mode

Explanation:

If a qos making (MPLS EXP) is changed in the MPLS network, it is also changed in the egress IP Packet—short-pipe mode

This provide QOS transparency where the customer QOS marking (for example,DSCP) in the IP Packet is preserved.The egress PE uses the original customer QOS marking instead of the service provider QOS marking—pipe mode

Qos is done on the output interface of the PE router that is based on the receive -----Uniform mode

QUESTION NO: 71

Scenario:

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on the router icon to gain access to the console of the router.

No console or enable passwords are required.

There are **four** multiple-choice questions with this task. Be sure to answer all **four** questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation.

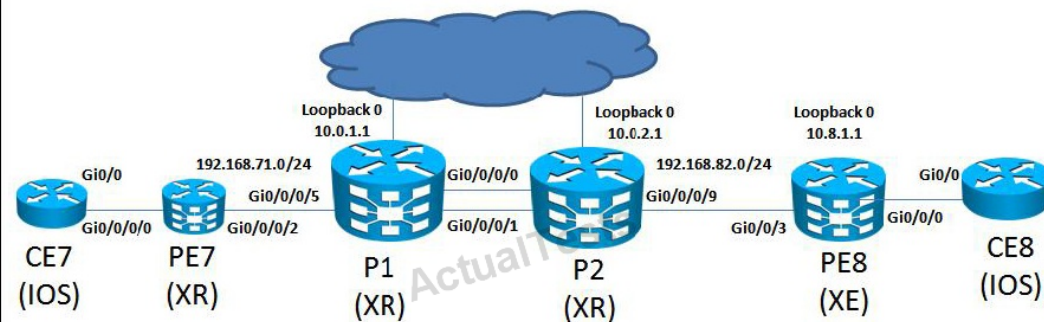
For example, the show running-config command is **NOT** supported in this simulation.

All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

Scenario

Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the PE8 and P2 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.

Topology



In this simulation, you will only have access to the PE8 and P2 consoles
Click on the PE8 and P2 router icon to access the respective console
PE8 is an ASR1K and P2 is an ASR9K



On the PE8-router what is the label value used to reach the 10.7.1.1/32 network prefix?

- A. 10.0.2.1:0
- B. imp-null
- C. 16024
- D. 128

Answer: A

Explanation: Need to run command at PE8 router so that we can get the value .Local tag value is answer for these

#show mpls forwarding-table 10.7.1.1

QUESTION NO: 72

Scenario:

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on the router icon to gain access to the console of the router.

No console or enable passwords are required.

There are **four** multiple-choice questions with this task. Be sure to answer all **four** questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation.

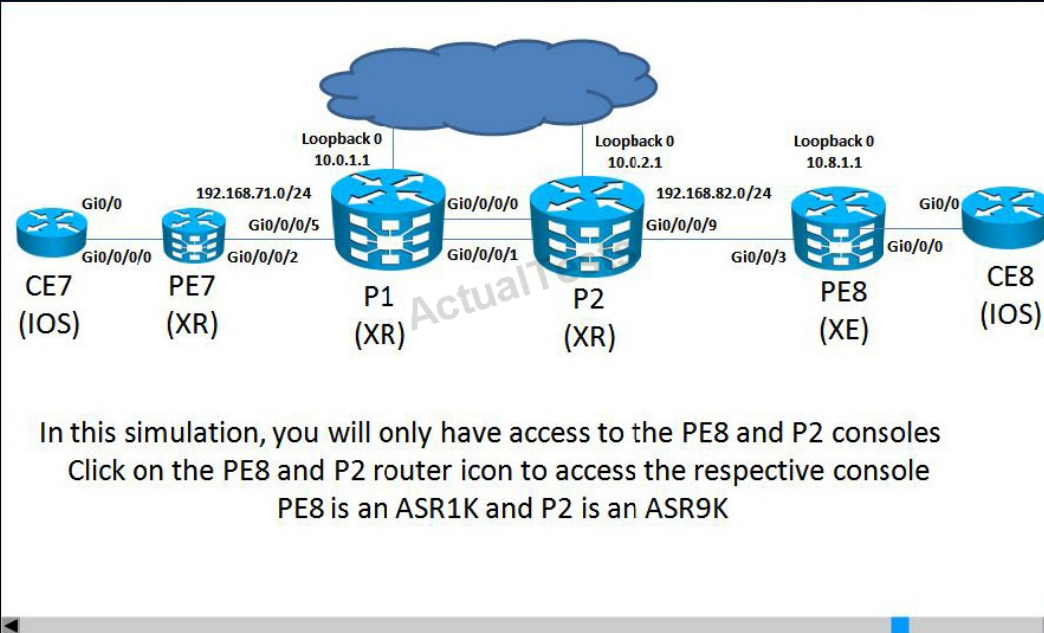
For example, the show running-config command is **NOT** supported in this simulation.

All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

Scenario

Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the PE8 and P2 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.

Topology





On P2IOS-XR router what is targeted LDP hellos hold time on LDP session with 10.0.1.1:0 LDP ID?

- A. 90 sec
- B. 10 sec
- C. 3 sec
- D. 8.6 sec

Answer: B

Explanation: Need to run command at PE8 router so that we can get the value .

#show mpls ldp parameters

show mpls ldp neighbor detail

QUESTION NO: 73

Scenario:

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on the router icon to gain access to the console of the router.

No console or enable passwords are required.

There are **four** multiple-choice questions with this task. Be sure to answer all **four** questions before selecting the Next button.

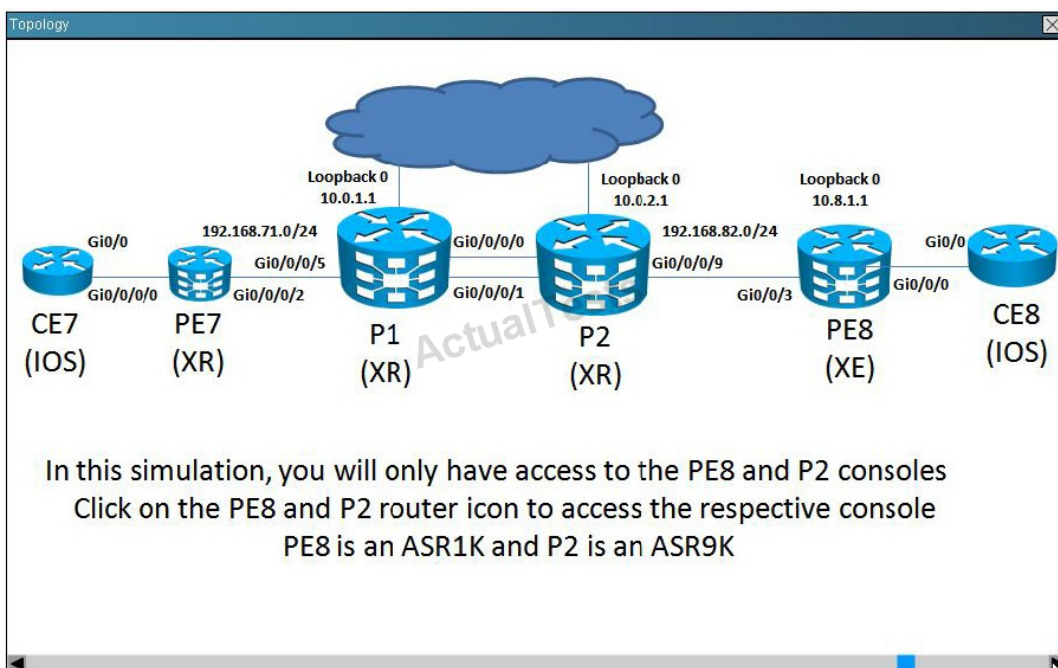
Not all the CLI commands or commands options are supported or required for this simulation.

For example, the show running-config command is **NOT** supported in this simulation.

All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

Scenario

Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the PE8 and P2 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.





From PE8 router, what is the label action used to reach the P2 loopback 0 interface 10.0.2.1/32?

- A. Swap label 35 with label 16004
- B. Push label 16004
- C. Pop label 35
- D. Pop label 16009
- E. Push Label 16009
- F. Swap label 35 with label 16009

Answer: E

Explanation:

Need to run command at PE8 router so that we can get the value .Local tag value is answer for these

#show mpls forwarding-table 10.0.2.1

QUESTION NO: 74

Scenario:

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on the router icon to gain access to the console of the router.

No console or enable passwords are required.

There are **four** multiple-choice questions with this task. Be sure to answer all **four** questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation.

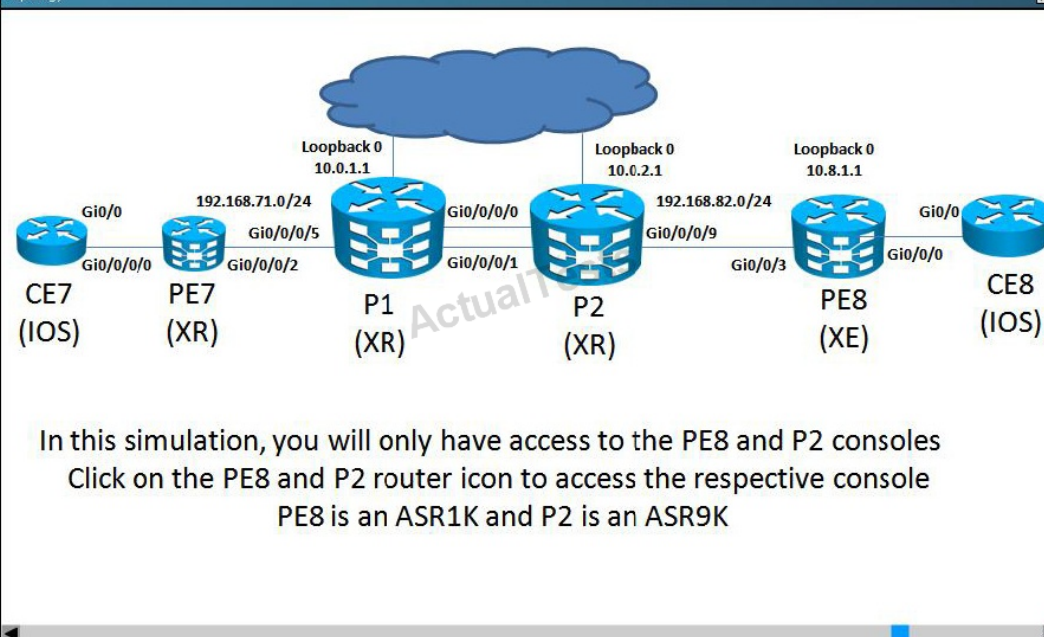
For example, the show running-config command is **NOT** supported in this simulation.

All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

Scenario

Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the PE8 and P2 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.

Topology





From the PE8 router, how many total prefixes for have an incoming local label assigned (including the imp-null)?

- A. 45
- B. 21
- C. 66
- D. 22

Answer: D

Explanation: Need to run command at PE8 router so that we can get the value .prefix value is answer for these

#show mpls forwarding-table

QUESTION NO: 75

Scenario:

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on the router icon to gain access to the console of the router.

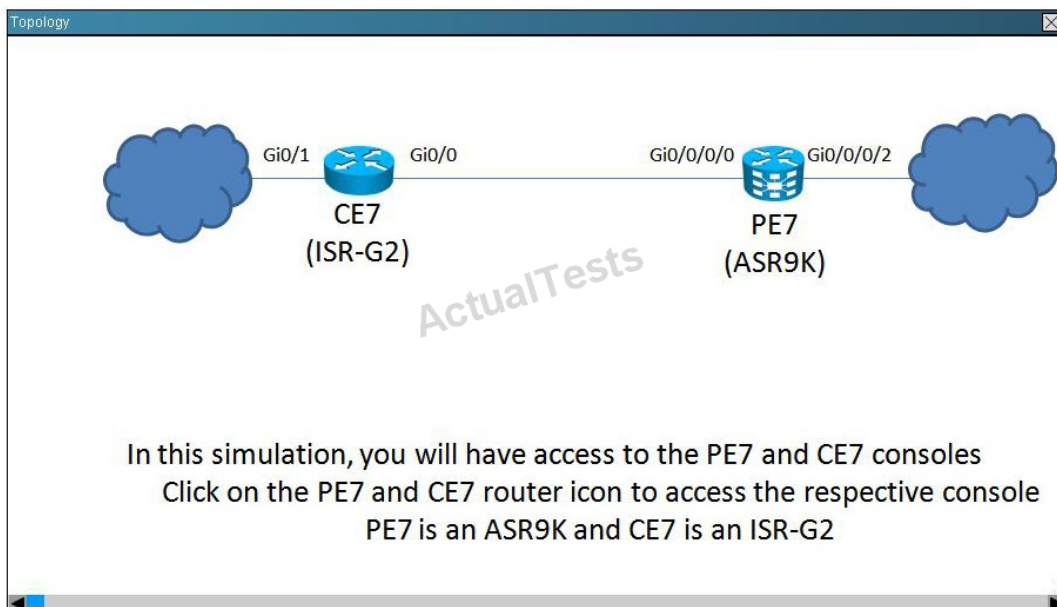
No console or enable passwords are required.

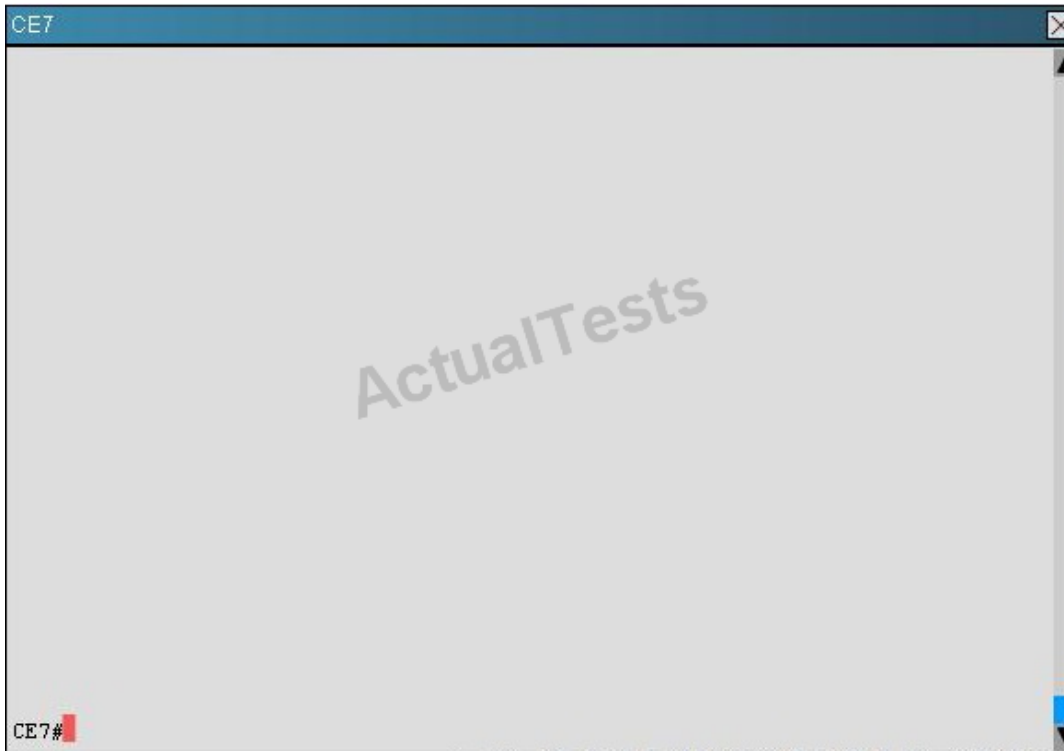
There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation. All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

Scenario

Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the CE7 and PE7 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.





On CE 7 router, which statement is correct regarding the "QOS-POLICY policy map configurations?

- A.** Traffic matched by the "QOS-HTTP-1" class-map is shaped to an average rate of 2560000 128000bps
- B.** Traffic matched by the "QOS1-HTTP-2" class-map will be queued in the low-latency-queue which has a maximum bandwidth guarantee of 64000
- C.** Traffic matched by the "QOS-FTP-1" class-map can't use more than 256 Kbps under any

condition

D. The "QOS-POLICY is applied to the gi0/0 interface in the input direction

Answer: C

Explanation: # show policy-map

QUESTION NO: 76

Scenario:

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on the router icon to gain access to the console of the router.

No console or enable passwords are required.

There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation. All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

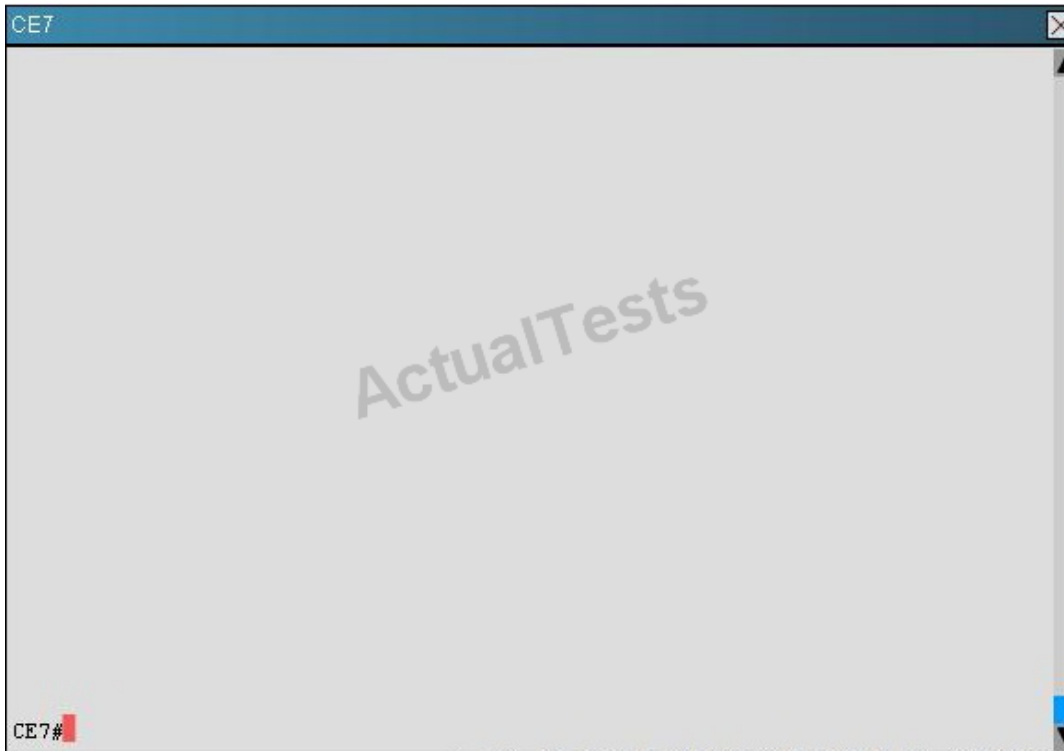
Scenario

Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the CE7 and PE7 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.

Topology



In this simulation, you will have access to the PE7 and CE7 consoles
Click on the PE7 and CE7 router icon to access the respective console
PE7 is an ASR9K and CE7 is an ISR-G2



After reviewing present router configuration CE7 .which two statements are correct regarding behavior of the "llq" policy-map? (Choose 2)

- A. Traffic matched by the "cisco1" class-map will be assigned to the priority queue.
- B. The "llq" QoS policy is applied to the gi0/0 interface in the output direction
- C. Traffic matched by the "cisco2" class-map has a maximum bandwidth of 30%
- D. Traffic matched by the "cisco3" class-map has no priority and has a minimum bandwidth guarantee of 20%

E. There are no packets being matched by the "class-default" traffic class in the "llq" policy

Answer: A,E

Explanation: #show policy-map interface <name>

QUESTION NO: 77

Scenario:

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on the router icon to gain access to the console of the router.

No console or enable passwords are required.

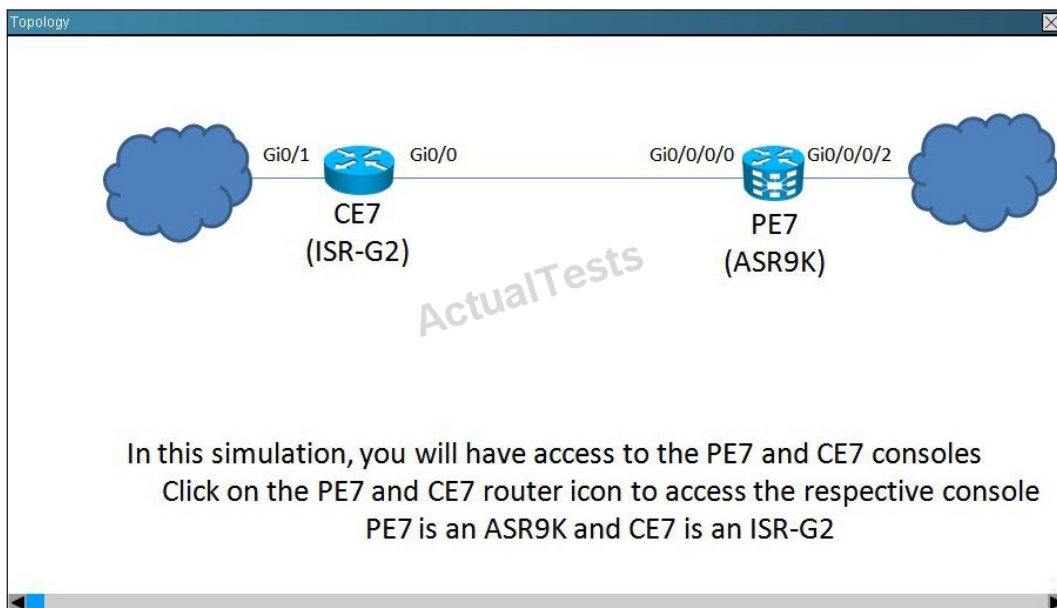
There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

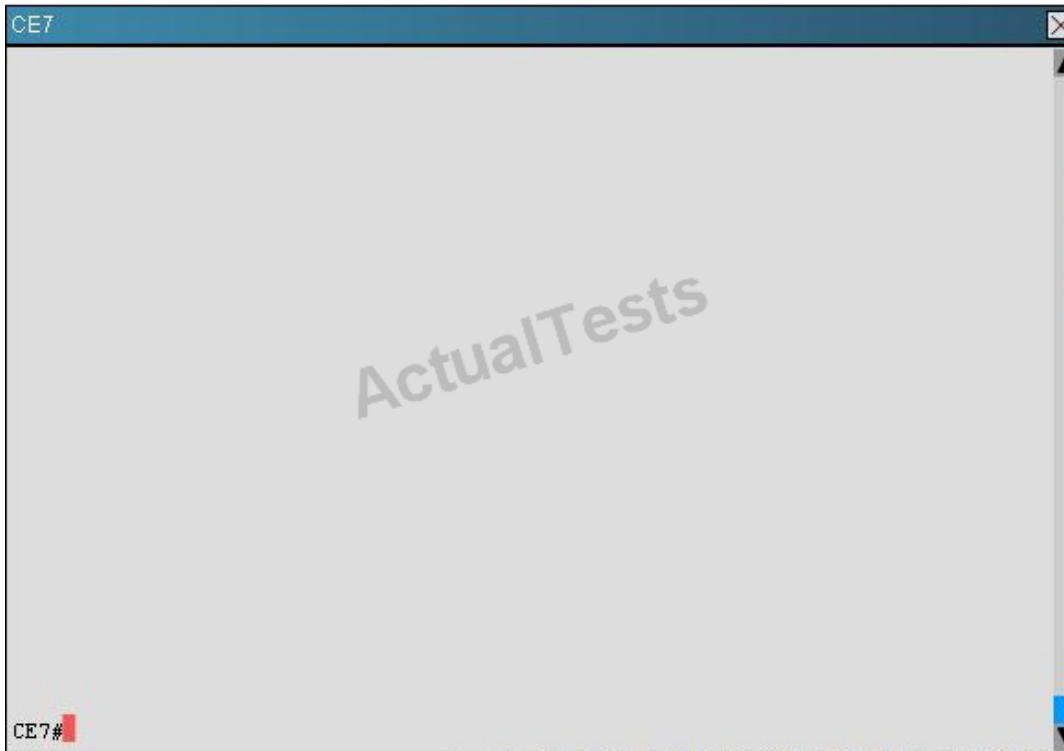
Not all the CLI commands or commands options are supported or required for this simulation.

All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

Scenario

Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the CE7 and PE7 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.





On CE7 which statement is correct regarding Traffic the QoS policy applied to the gi0/0 interface in the input direction?

- A. Traffic matching the "MARKING2" class-map will be marked with qos-group 5
- B. Traffic matching the "MARKING1" class-map will be marked with MPLS EXP 5
- C. All incoming packets not matched by the "MARKING1" class-map will be dropped
- D. All incoming packets with the DSCP EF marking will be marked with MPLS EXP 0

Answer: A

Explanation: #show policy-map interface <name> ---check the class-map field

QUESTION NO: 78

Scenario:

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on the router icon to gain access to the console of the router.

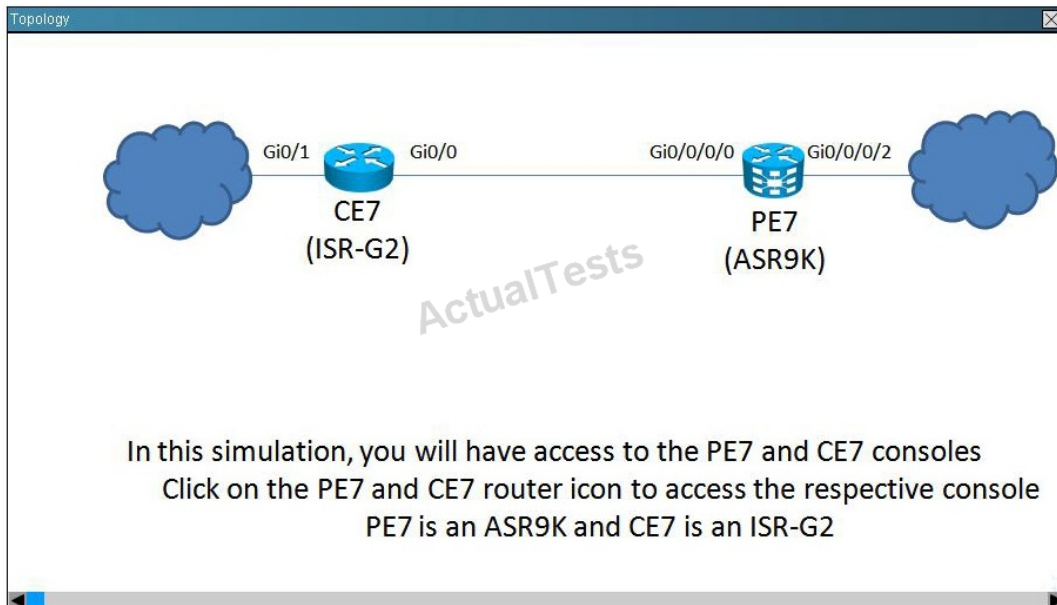
No console or enable passwords are required.

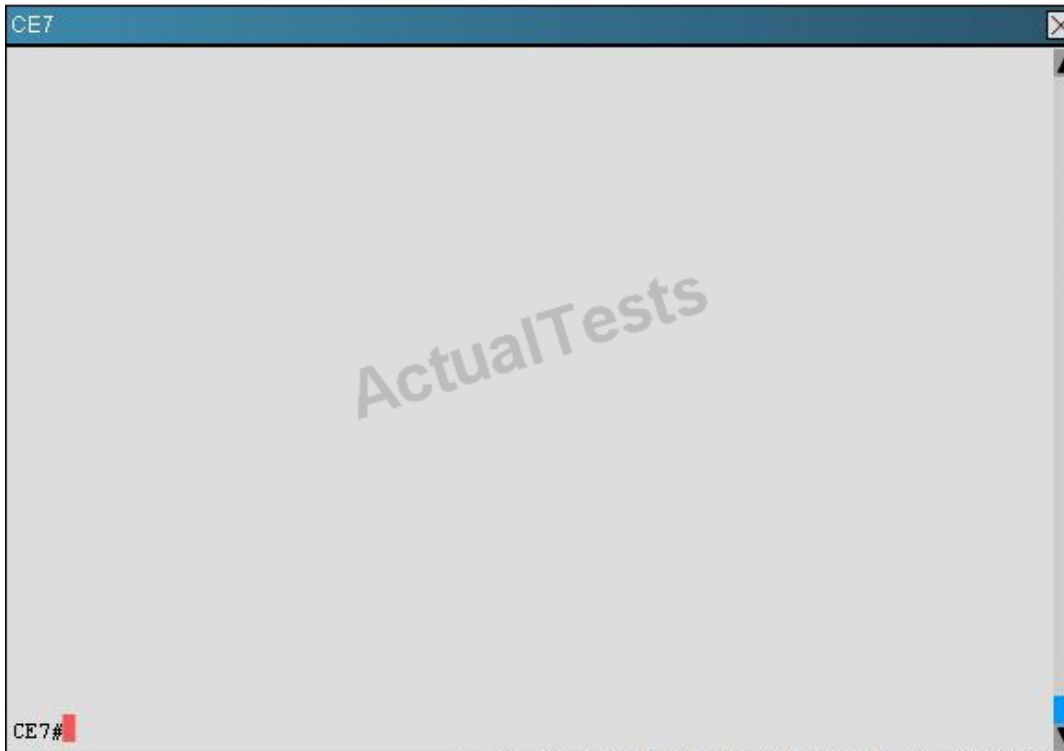
There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation. All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

Scenario

Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the CE7 and PE7 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.





On PE7 which statement is correct regarding the "traffic-policing" policy-map?

- A. The "traffic-policing" QoS policy is applied to the Gi0/0/0/0 interface in the outbound direction.
- B. The "traffic-policing" QoS policy will police the traffic matched by the "test12" ACL to 3 mbps and will drop all exceeding traffic.
- C. All incoming traffic not matched by the "test12" ACL to the Gi0/0/0/0 interface will be dropped.
- D. The "traffic-policing" QoS policy is using dual rate class-based traffic policing.

Answer: B

Explanation: #show policy-map